Zhuorui Zhao

PhD Candidate | AI for Hardware Design and Verification

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Researcher in electronic design automation with experience building LLM workflows for Verilog generation, testbench generation, and system-level verification. Also worked on FPGA acceleration and AI chip deployment.

ISQED 2025 Best Paper Award for VRank, a first-author paper on Verilog generation.

Selected Research

VRank | ISQED 2025

- Developed a workflow that groups Verilog candidates by simulated behavior and ranks them by self-consistency. On VerilogEval-Human, functional correctness (pass@1) improved by 10.5% on average across multiple LLMs.

VFocus | SOCC 2025

- Designed a three-stage workflow for candidate filtering, simulation-based ranking, and targeted reasoning over inconsistent outputs. Improved pass@1 across multiple reasoning models on VerilogEval-Human.

VClare | NeurIPS 2026 Workshop (accepted)

- Developed a specification repair framework and two benchmarks for incomplete, ambiguous, or contradictory hardware specifications. The paper reports Verilog design pass-rate gains of 12.7% on single-module tasks and 13.7% on multi-module tasks.

Education

Technical University of Munich | Mar 2022 - Present

- PhD candidate in Computer Engineering. Advisor: Prof. Ulf Schlichtmann.

University of Notre Dame | Sep 2020 - Jan 2022

- M.S., Computer Science and Engineering. GPA: 3.90/4.00. Advisor: Prof. Yiyu Shi.

Beijing Institute of Technology | Sep 2016 - Jun 2020

- B.Eng., Electronic Information Engineering. GPA: 3.70/4.00.

Technical Skills

Languages: Python, C, C++, MATLAB, Verilog, BangC

Hardware and AI: Vivado, Vitis HLS, FPGA deployment, RTL simulation, testbench generation, LLM agents

Research & Engineering Experience

Technical University of Munich, EDA Lab | Mar 2022 - Present

- Research Assistant. Built and evaluated LLM workflows for RTL generation and verification, including simulation-based candidate selection and specification repair.

Zhejiang University, Prof. Cheng Zhuo's Lab | Oct 2025 - Mar 2026

- Visiting Student. Combined VRank with testbench generation to improve expected-output generation and contributed to ReflectBench with collaborators from Shandong University.

University of Notre Dame | Jul 2019 - Jan 2022

- Research Assistant. Used transfer learning to speed up neural architecture search by about 3x and explored reinforcement-learning-based search for binary neural networks.
- Worked with EdgeCortix to accelerate MobileNet on FPGA using Vitis HLS and Python, improving performance through pipelining and fewer memory writes.
- Developed a contrastive-learning model for RGB-D classification, achieving state-of-the-art results on SUN-RGBD and Washington RGB-D.

Cambricon Technologies | Aug 2019 - Oct 2019

- Software Intern. Deployed ResNet-50 and Inception-v3 on the MLU100 AI accelerator and implemented vector operations in BangC.

Beijing Institute of Technology | May 2019 - Aug 2019

- Undergraduate Intern. Led model training and architecture search for an FPGA-based license plate recognition system that won national second prize in the ARM Cup.

Tsinghua University, Institute of Microelectronics | Aug 2018 - Jul 2019

- Research Assistant. Helped design cache access for a CGRA processing unit, fixed critical bugs, and implemented UART and APB interfaces.
- Proposed a network-on-chip memory design that increased memory bandwidth by about 10x.

Additional Publications

ReflectBench | DAC 2026. Co-author. System-level testbench generation through consensus and reflection.

Large Language Models (LLMs) for Electronic Design Automation (EDA) | SOCC 2025 special session. Co-author.

LLM-Aided Efficient Hardware Design Automation | arXiv, 2024. Co-author.

Additional Honors

Outstanding Graduate, Beijing Institute of Technology and Beijing Municipality (2020)

National Second Prize, ARM Cup, National Undergraduate Integrated Circuit Innovation and Entrepreneurship Competition (2019)